

CO1-AT1 - Short Answers

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COURSE CODE: ITA0508

COURSE NAME: Computer Vision

QUESTION 1

```
[7]
✓ 0s
import cv2
import matplotlib.pyplot as plt
from google.colab import files

[8]
✓ 1m
uploaded = files.upload()

... Choose files sample.jpg
sample.jpg(image/jpeg) - 102967 bytes, last modified: 21/07/2026 - 100% done
Saving sample.jpg to sample.jpg

[9]
✓ 2s
filename = list(uploaded.keys())[0]

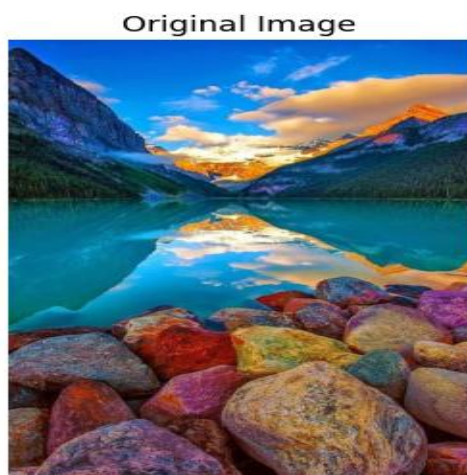
image = cv2.imread(filename)

image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")
plt.show()

blurred = cv2.GaussianBlur(image, (25,25), 0)
plt.imshow(blurred)
plt.title("Blurred Image")
plt.axis("off")
plt.show()

cv2.imwrite("blurred_output.jpg", cv2.cvtColor(blurred, cv2.COLOR_RGB2BGR))
files.download("blurred_output.jpg")
```

Output:



QUESTION 2

```
[10] ▶ |pip install opencv-python
import cv2
import matplotlib.pyplot as plt
from google.colab import files
# Upload image
uploaded = files.upload()
# Read image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
# Display original image
plt.figure(figsize=(6,6))
plt.imshow(image)
plt.title("Original Low-Light Image")
plt.axis("off")
plt.show()
# Enhance brightness and contrast
enhanced = cv2.convertScaleAbs(image, alpha=1.6, beta=40)
# Display enhanced image
plt.figure(figsize=(6,6))
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")
plt.show()
# Save output
cv2.imwrite("enhanced_output.jpg", cv2.cvtColor(enhanced, cv2.COLOR_RGB2BGR))
# Download output
files.download("enhanced_output.jpg")
```

Output:

Original Low-Light Image



Enhanced Image



QUESTION 3

```
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow
uploaded = files.upload()
img = cv2.imread("surveillance.jpg")

if img is None:
    print("Image not found!")
else:
    print("Image uploaded successfully.")
    print("Original Surveillance Image")
    cv2_imshow(img)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    print("Grayscale Image")
    cv2_imshow(gray)
```

Output:



QUESTION 4

```
!pip install opencv-python

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

filename = list(uploaded.keys())[0]

image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Brightness and contrast enhancement
enhanced = cv2.convertScaleAbs(image, alpha=1.5, beta=40)

# Display original
plt.figure(figsize=(6,6))
plt.imshow(image)
plt.title("Original Road Image")
plt.axis("off")
plt.show()

# Display enhanced
plt.figure(figsize=(6,6))
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")
plt.show()
```

Output:

Original Road Image



Enhanced Image



QUESTION 5

```
# Install OpenCV
!pip install opencv-python
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Noise reduction
denoised = cv2.GaussianBlur(image, (5,5), 0)

# Display original image
plt.figure(figsize=(6,6))
plt.imshow(image)
plt.title("Noisy Image")
plt.axis("off")
plt.show()

# Display denoised image
plt.figure(figsize=(6,6))
plt.imshow(denoised)
plt.title("Noise Reduced Image")
plt.axis("off")
plt.show()

# Save output
cv2.imwrite("denoised_output.jpg", cv2.cvtColor(denoised, cv2.COLOR_RGB2BGR))

# Download output
files.download("denoised_output.jpg")
```

Output:

Noisy Image



Noise Reduced Image

